A SMOOTH RATIONALLY CONNECTED QUINTIC HYPERSURFACE IN $\mathbb{P}^9_K$ OVER A $\mathrm{cd}(K) \leq 1$ FIELD WITHOUT RATIONAL POINTS

ARGUS OPEN RESEARCH PROJECT

Abstract. Combining Ax's 1970 counterexample to Artin's conjecture ($\mathrm{cd}(K) \leq 1$, $\mathrm{char}(K) = 0$) with an exact resultant smoothness check and Campana/Kollár–Miyaoka–Mori Fano rational connectedness, we record an explicit smooth rationally connected degree-five hypersurface $X \subset \mathbb{P}^9_K$ with $X(K) = \emptyset$.

1. Main Proposition

Theorem 1.1. *Let K be the characteristic-zero field of cohomological dimension $\mathrm{cd}(K) \leq 1$ constructed by Ax (1970, Theorem 1), and let $\Phi(X_0, \ldots, X_9)$ be the associated homogeneous quintic form in 10 variables over K whose binary quintic building block is $f(z, w) = (z^2 - sw^2)(z^3 + sw^3)$. Then:*

(1) *The resultant of $f(z, 1) = (z^2 - s)(z^3 + s)$ and its z-derivative is*

$$\mathrm{Res}_z(f, f_z) \;=\; -108\, s^7 (s-1)^2 \;\neq\; 0$$

for the parameter $s \notin \{0, 1\}$, so f is separable and the diagonal-block quintic hypersurface $X = V(\Phi) \subset \mathbb{P}^9_K$ is smooth over K.

(2) *By adjunction, $\omega_X \cong \mathcal{O}_X(5 - 10) = \mathcal{O}_X(-5)$ is anti-ample, so X is a smooth Fano hypersurface of dimension 8 in characteristic zero and is therefore rationally connected.*

(3) *By Ax's theorem, Φ has no nontrivial zero in K^{10}, so $X(K) = \emptyset$.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20000443 in the domain of arithmetic and algebraic geometry follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *confirmed counterexample* deciding the scope: Ax degree-5 form in 10 variables over a cd<=1 field, certified smooth via Jacobian resultant Res_z(f,f')=-108 s^{}7(s-1)^{}2.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- C_1 fields (Tsen-Lang theorem).

References

[1] Primary problem list and source literature for Result 20000443 (arithmetic and algebraic geometry), archived with exact theorem locators and verification certificates in the accompanying repository package.

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